

Machine Learning: Model Evaluation Basics

Synthetic educational material designed for testing extraction, comparison, and reasoning.

1. What Is Model Evaluation?

Model evaluation measures how well a machine-learning model performs on data that represents the situations in which it will be used. Evaluation should distinguish training performance from generalization performance.

2. Common Classification Metrics

Metric	Interpretation
Accuracy	Fraction of predictions that are correct overall.
Precision	Among predicted positives, the fraction that are actually positive.
Recall	Among actual positives, the fraction detected by the model.
F1 score	Harmonic mean of precision and recall.
Specificity	Among actual negatives, the fraction correctly identified as negative.

3. Train, Validation, and Test Sets

A training set is used to fit model parameters. A validation set can support model selection and hyperparameter tuning. A final test set should be kept separate until the evaluation stage so that the reported result is less likely to be optimized indirectly against it.

4. Overfitting

Overfitting occurs when a model learns patterns that work very well on its training data but fail to generalize. Warning signs include a large gap between training and validation performance. Possible responses include collecting more representative data, simplifying the model, regularization, early stopping, or improving the data pipeline.

5. Example Interpretation

Suppose a classifier has high accuracy but very low recall for an important positive class. The accuracy alone may hide a serious weakness. In such a situation, recall, precision, the confusion matrix, and the consequences of false negatives should be examined before deciding that the model is effective.